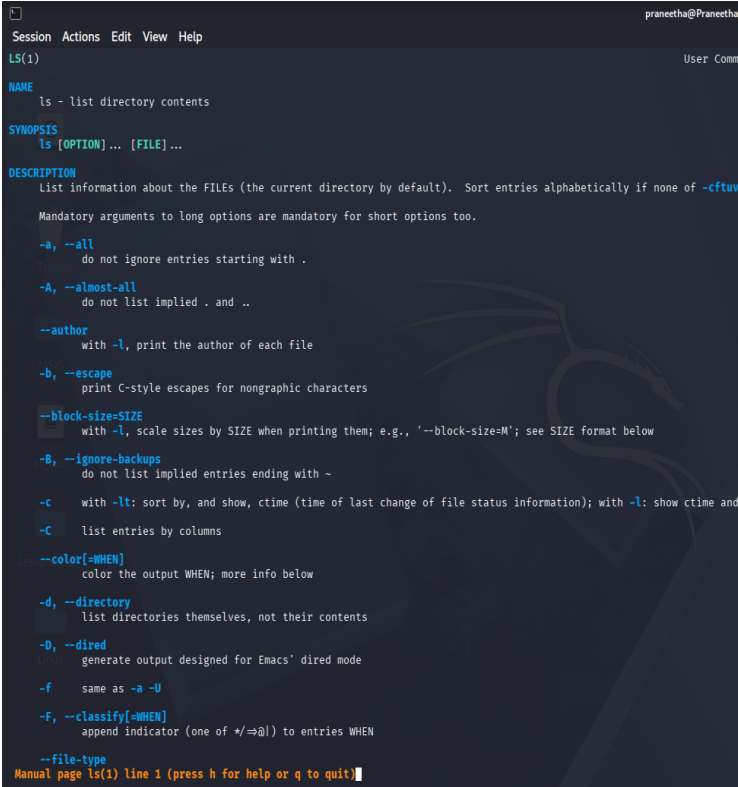


The main aim of this lab session is to provide hands-on experience on

- Getting Help
- Basic Commands
- Navigation
- File System
- Simple shell script
- Globbing
- Extended Globbing
- Filters -I (head, tail, wc)

1. Getting Help

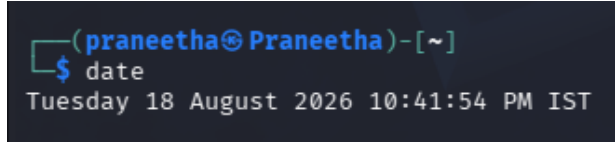
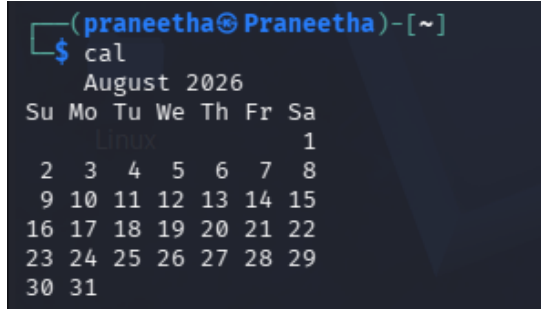
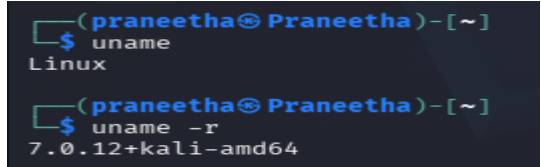
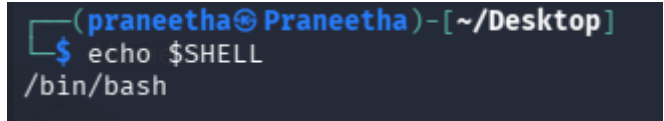
Task	Command Name	Syntax	Example	Screenshots
To get manual page for the known command	man	man <command>	man ls	 A screenshot of a terminal window showing the manual page for the 'ls' command. The window title is 'praneetha@Praneetha'. The menu bar includes 'Session', 'Actions', 'Edit', 'View', and 'Help'. The content is titled 'LS(1)' and 'User Commands'. It includes sections for NAME, SYNOPSIS, DESCRIPTION, and various options like -a, -A, --author, -b, --block-size=SIZE, -B, --ignore-backups, -c, -C, --color[=WHEN], -d, --directory, -D, --dired, -f, -F, --classify[=WHEN], and --file-type. The bottom of the screen shows 'Manual page ls(1) line 1 (press h for help or q to quit)'.

To get manual page for the unknown command	man -k	man -k <keyword>	man -k copy	<pre>(praneetha@Praneetha)-[~/Desktop] \$ man -k copy Clone (3pm) - recursively copy Perl datatypes bcopy (3) - copy byte sequence cifsd (8) - convert and copy a file over SMB copy (7) - copy data between a file and a table copy_file_range (2) - Copy a range of data from one file to another copyfilerange (1) - copy range(s) from source to destination file copysign (3) - copy sign of a number copysignf (3) - copy sign of a number copysignl (3) - copy sign of a number cp (1) - copy files and directories cpio (1) - copy files to and from archives dd (1) - convert and copy a file debconf-copydb (1) - copy a debconf database dvi (1) - produce modified copy of DVI file getunwind (2) - copy the unwind data to caller's buffer getutmp (3) - copy utmp structure to utmpx, and vice versa getutmpx (3) - copy utmp structure to utmpx, and vice versa git-checkout-index (1) - Copy files from the index to the working tree i686-w64-mingw32-objcopy (1) - copy and translate object files install (1) - copy files and set attributes llvm-dwarfutil-21 (1) - A tool to copy and manipulate debug info llvm-objcopy-21 (1) - object copying and editing tool mariadb-hotcopy (1) - a database backup program (mariadb-hotcopy is now a symlink) memccpy (3) - copy memory area memcpy (3) - copy memory area memmove (3) - copy memory area mempcpy (3) - copy memory area ntfs (8) - copy file to an NTFS volume. objcopy (1) - copy and translate object files</pre>
To know the source file binary	whereis	whereis <command>	whereis cat	<pre>(praneetha@Praneetha)-[~] \$ whereis cat cat: /usr/bin/cat /usr/share/man/man1/cat.1.gz</pre>
To know the path of the command	which	which <command>	which cat	<pre>(praneetha@Praneetha)-[~] \$ which cat /usr/bin/cat</pre>
To know the command is external or internal	type	type <command>	type pwd (internal) type mkdir (external)	<pre>(praneetha@Praneetha)-[~] \$ type pwd pwd is a shell builtin (praneetha@Praneetha)-[~] \$ type mkdir mkdir is /usr/bin/mkdir</pre>

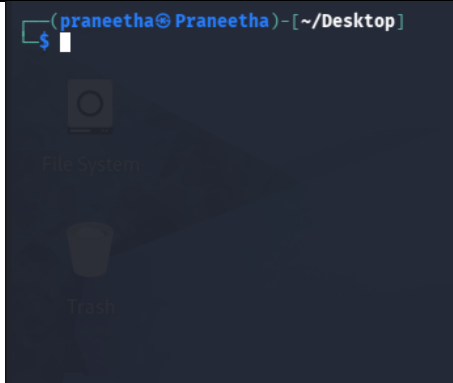
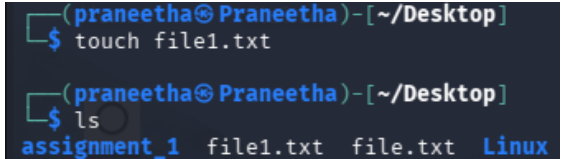
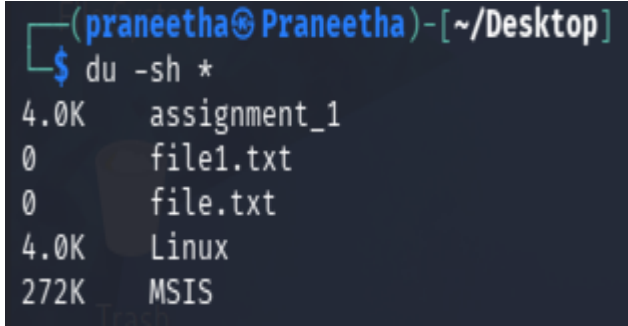
To get help for the internal command	help	help <command>	help pwd	<pre> (praneetha@Praneetha)-[~] \$ help pwd pwd: pwd [-LP] Print the name of the current working directory. Options: -L print the value of \$PWD if it names the current working directory -P print the physical directory, without any symbolic links links By default, 'pwd' behaves as if '-L' were specified. Exit Status: Returns 0 unless an invalid option is given or the current directory cannot be read. </pre>
To list out bash commands	compgen	compgen -b	compgen -b	<pre> (praneetha@Praneetha)-[~] \$ compgen -b . : [alias bg bind break builtin caller cd command compgen complete compgen continue declare dirs disown echo enable eval exec exit export false fc fg getopts hash help history jobs kill </pre>

To know the usage of the command	--help	<command> --help	pwd --help	<pre> (praneetha@Praneetha)-[~] \$ pwd --help pwd: pwd [-LP] Print the name of the current working directory. Options: -L print the value of \$PWD if it names the current working directory -P print the physical directory, without any symbolic links -h, --help display this help message By default, `pwd' behaves as if `-L' were specified. Exit Status: Returns 0 unless an invalid option is given or the current directory cannot be read. </pre>
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2. Basic Commands

Task	Command Name	Syntax	Example	Screenshots
To know today's date	date	date	date	 <pre>(praneetha@Praneetha)-[~] \$ date Tuesday 18 August 2026 10:41:54 PM IST</pre>
To print calendar	cal	cal	cal	 <pre>(praneetha@Praneetha)-[~] \$ cal August 2026 Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31</pre>
To print kernel version	uname	uname -r	uname -r	 <pre>(praneetha@Praneetha)-[~] \$ uname Linux (praneetha@Praneetha)-[~] \$ uname -r 7.0.12+kali-amd64</pre>
To print default shell	echo	echo \$SHELL	echo \$SHELL	 <pre>(praneetha@Praneetha)-[~/Desktop] \$ echo \$SHELL /bin/bash</pre>

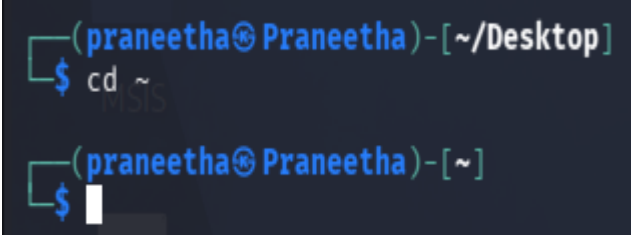
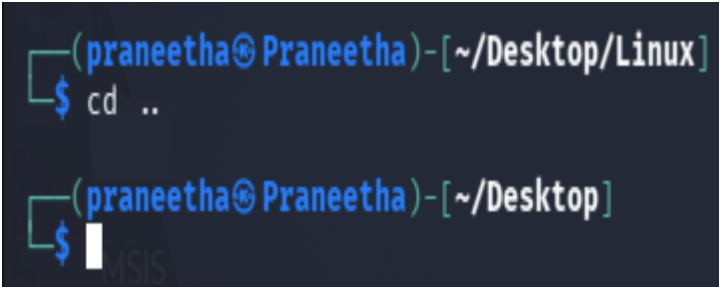
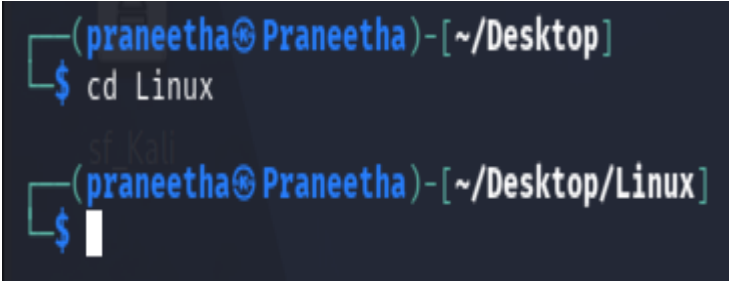
To print currently logged in user	whoami	whoami	whoami	<pre>(praneetha@Praneetha)-[~/Desktop] \$ whoami praneetha</pre>
To create shortcut for command	alias	alias <name> = '<command>'	alias ll = 'ls -l'	<pre>(praneetha@Praneetha)-[~/Desktop] \$ alias ll='ls -l' (praneetha@Praneetha)-[~/Desktop] \$ alias ll alias ll='ls -l'</pre>
To delete shortcut	unalias	unalias <name>	unalias ll	<pre>(praneetha@Praneetha)-[~/Desktop] \$ unalias ll (praneetha@Praneetha)-[~/Desktop] \$ alias ll bash: alias: ll: not found</pre>
To change the timestamp of the file	touch	touch <filename>	touch file.txt	<pre>(praneetha@Praneetha)-[~/Desktop] \$ touch file.txt (praneetha@Praneetha)-[~/Desktop] \$ ls -l total 12 drwxrwxr-x 2 praneetha praneetha 4096 Aug 13 18:37 assignment_1 -rw-rw-r-- 1 praneetha praneetha 0 Aug 18 23:05 file.txt</pre>
To clear the screen	clear	clear	clear	<pre>(praneetha@Praneetha)-[~/Desktop] \$ clear</pre>

				
To create empty files	touch	touch <filename>	touch file1.txt	
To know disk usage	du	du -sh *	du -sh *	

To know free space in the system	df	df -h	df -h	<pre>(praneetha@Praneetha) - [~/Desktop] \$ df -h Filesystem Size Used Avail Use% Mounted on udev 1.9G 0 1.9G 0% /dev tmpfs 392M 996K 391M 1% /run /dev/sda1 38G 18G 18G 50% / tmpfs 2.0G 4.0K 2.0G 1% /dev/shm none 1.0M 0 1.0M 0% /run/credentials/systemd-journald.service tmpfs 2.0G 8.0K 2.0G 1% /tmp Kali 202G 191G 11G 95% /media/sf_Kali none 1.0M 0 1.0M 0% /run/credentials/getty@tty1.service tmpfs 392M 108K 392M 1% /run/user/1000</pre>
To know about the Linux release	cat	cat /etc/os-release	cat /etc/os-release	<pre>(praneetha@Praneetha) - [~/Desktop] \$ cat /etc/os-release PRETTY_NAME="Kali GNU/Linux Rolling" NAME="Kali GNU/Linux" VERSION_ID="2026.3" VERSION="2026.3" VERSION_CODENAME=kali-rolling ID=kali ID_LIKE=debian HOME_URL="https://www.kali.org/" SUPPORT_URL="https://forums.kali.org/" BUG_REPORT_URL="https://bugs.kali.org/" ANSI_COLOR="1;31"</pre>
To show login history	last	last	last	<pre>(praneetha@Praneetha) - [~/Desktop] \$ last praneeth tty7 :0 Tue Aug 18 22:52 - still logged in praneeth tty7 :0 Tue Aug 18 20:00 - still logged in praneeth tty7 :0 Thu Aug 13 18:18 - still logged in praneeth tty7 :0 Tue Aug 11 15:19 - still logged in praneeth tty7 :0 Mon Aug 10 22:19 - still logged in praneeth tty7 :0 Thu Jul 30 23:57 - 00:14 (00:17) wtm pdb begins Thu Jul 30 23:57:20 2026</pre>

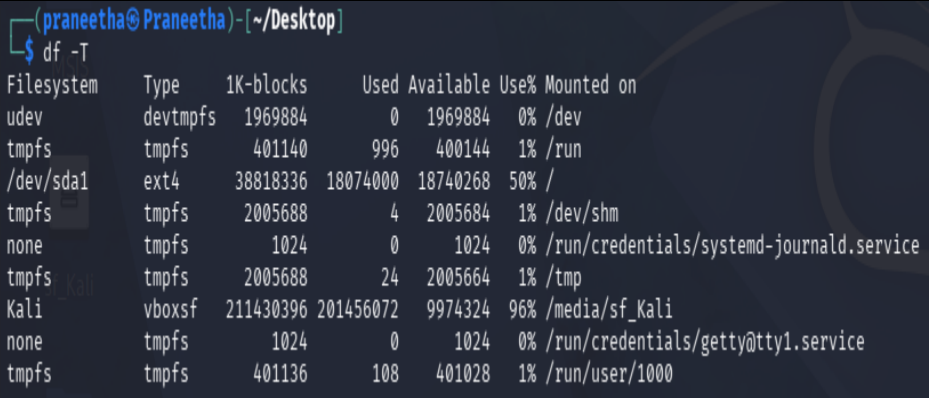
To check the current user privileges	sudo	sudo -l	sudo -l	<pre>(praneetha@Praneetha)-[~/Desktop] \$ sudo -l Matching Defaults entries for praneetha on Praneetha: secure_path=/usr/sbin\:usr/bin\:sbin\:bin Runas and Command-specific defaults for praneetha: Defaults!/usr/sbin/visudo env_keep+="SUDO_EDITOR EDITOR VISUAL" User praneetha may run the following commands on Praneetha: (ALL : ALL) ALL</pre>
To find about the Linux release	lsb_release	lsb_release -a	lsb_release -a	<pre>(praneetha@Praneetha)-[~/Desktop] \$ lsb_release -a No LSB modules are available. Distributor ID: Kali Description: Kali GNU/Linux Rolling Release: 2026.3 Codename: kali-rolling</pre>
To check system uptime	uptime	uptime	uptime	<pre>(praneetha@Praneetha)-[~/Desktop] \$ uptime 23:29:02 up 36 min, 1 user, load average: 0.32, 0.53, 0.61</pre>
To shutdown	shutdown	sudo shutdown now	sudo shutdown now	<pre>(praneetha@Praneetha)-[~/Desktop] \$ sudo shutdown now</pre>
To restart	reboot	sudo reboot	sudo reboot	<pre>(praneetha@Praneetha)-[~/Desktop] \$ sudo reboot</pre>

3. Navigation

Task	Command	Syntax	Screenshots
To navigate home directory	cd	cd ~	 <pre>(praneetha@Praneetha)-[~/Desktop] \$ cd ~ (praneetha@Praneetha)-[~]</pre>
To navigate to the parent directory	cd	cd ..	 <pre>(praneetha@Praneetha)-[~/Desktop/Linux] \$ cd .. (praneetha@Praneetha)-[~/Desktop]</pre>
To navigate to the child directory	cd	cd <directory>	 <pre>(praneetha@Praneetha)-[~/Desktop] \$ cd Linux (praneetha@Praneetha)-[~/Desktop/Linux]</pre>

Alternate command to cd	pushd	pushd <directory>	<pre> (praneetha@Praneetha)-[~/Desktop] \$ pushd Linux ~/Desktop/Linux ~/Desktop assignment 1 (praneetha@Praneetha)-[~/Desktop/Linux] \$ </pre>
To go back to the previous directory	cd	cd -	<pre> (praneetha@Praneetha)-[~/Desktop/Linux] \$ cd - /home/praneetha/Desktop (praneetha@Praneetha)-[~/Desktop] \$ </pre>
To go to the root directory	cd	cd /	<pre> (praneetha@Praneetha)-[~/Desktop/Linux] \$ cd / (praneetha@Praneetha)-[/] \$ </pre>

4. File System

Task	Syntax	Command	Screenshot
How to identify the file system	df	df -T	 <pre> (praneetha@Praneetha) - [~/Desktop] \$ df -T Filesystem Type 1K-blocks Used Available Use% Mounted on udev devtmpfs 1969884 0 1969884 0% /dev tmpfs tmpfs 401140 996 400144 1% /run /dev/sda1 ext4 38818336 18074000 18740268 50% / tmpfs tmpfs 2005688 4 2005684 1% /dev/shm none tmpfs 1024 0 1024 0% /run/credentials/systemd-journald.service tmpfs tmpfs 2005688 24 2005664 1% /tmp Kali vboxsf 211430396 201456072 9974324 96% /media/sf_Kali none tmpfs 1024 0 1024 0% /run/credentials/getty@tty1.service tmpfs tmpfs 401136 108 401028 1% /run/user/1000 </pre>

5. Globbing

Create a dedicated practice directory containing sample files such as:

index.html, login.html, admin.htm, report.txt, access.log, error.log, user1.txt, user2.log, file123, backup1.html, backup2.txt

Use globbing patterns wherever possible, rather than specifying filenames individually.

- a. List all files ending in .html.

```
ls *.html
```

- b. List all files other than .html files using extended globbing.

```
ls !(*.html)
```

- c. List filenames containing at least one numeric digit.

```
ls *[0-9]*
```

- d. List files whose names begin with user followed by a single digit.

```
ls user[0-9].*
```

- e. List files ending in either .txt or .log using a single extended glob pattern.

```
ls *@(*.txt|*.log)
```

- f. List files that do not end in .txt or .log.

```
ls !(*.txt|*.log)
```

- g. List files beginning with either access or error using extended globbing.

```
ls @(access|error)*
```

- h. List .html files whose filenames contain a numeric digit.

```
ls *[0-9]*.html
```

```
(praneetha@Praneetha) - [~/Desktop]
$ mkdir practice

(praneetha@Praneetha) - [~/Desktop]
$ cd practice

(praneetha@Praneetha) - [~/Desktop/practice]
$ touch index.html login.html admin.htm report.txt access.log error.log user1.txt user2.log file123 backup1.html backup2.txt

(praneetha@Praneetha) - [~/Desktop/practice]
$ shopt -s extglob

(praneetha@Praneetha) - [~/Desktop/practice]
$ ls *.html
backup1.html index.html login.html

(praneetha@Praneetha) - [~/Desktop/practice]
$ ls !(*.html)
access.log admin.htm backup2.txt error.log file123 report.txt user1.txt user2.log

(praneetha@Praneetha) - [~/Desktop/practice]
$ ls *[0-9]*
backup1.html backup2.txt file123 user1.txt user2.log

(praneetha@Praneetha) - [~/Desktop/practice]
$ ls user[0-9].*
user1.txt user2.log

(praneetha@Praneetha) - [~/Desktop/practice]
$ ls *@(*.txt|*.log)
access.log backup2.txt error.log report.txt user1.txt user2.log

(praneetha@Praneetha) - [~/Desktop/practice]
$ ls !(*.txt|*.log)
admin.htm backup1.html file123 index.html login.html

(praneetha@Praneetha) - [~/Desktop/practice]
$ ls @(access|error)*
access.log error.log

(praneetha@Praneetha) - [~/Desktop/practice]
$ ls *[0-9]*.html
backup1.html
```

6. Extended Globbing

a. Assume a directory contains:

access.log, error.log, auth.log, auth.log.1, auth.log.2, system.log, notes.txt

- Write an extended glob pattern that selects only access.log, error.log, and auth.log.

```
ls @(access|error|auth).log
```

- Select all files except notes.txt.

```
ls !(notes.txt)
```

- Select files named either auth.log.1 or auth.log.2 using pattern matching rather than writing both complete filenames separately.

```
ls auth.log.@(1|2)
```

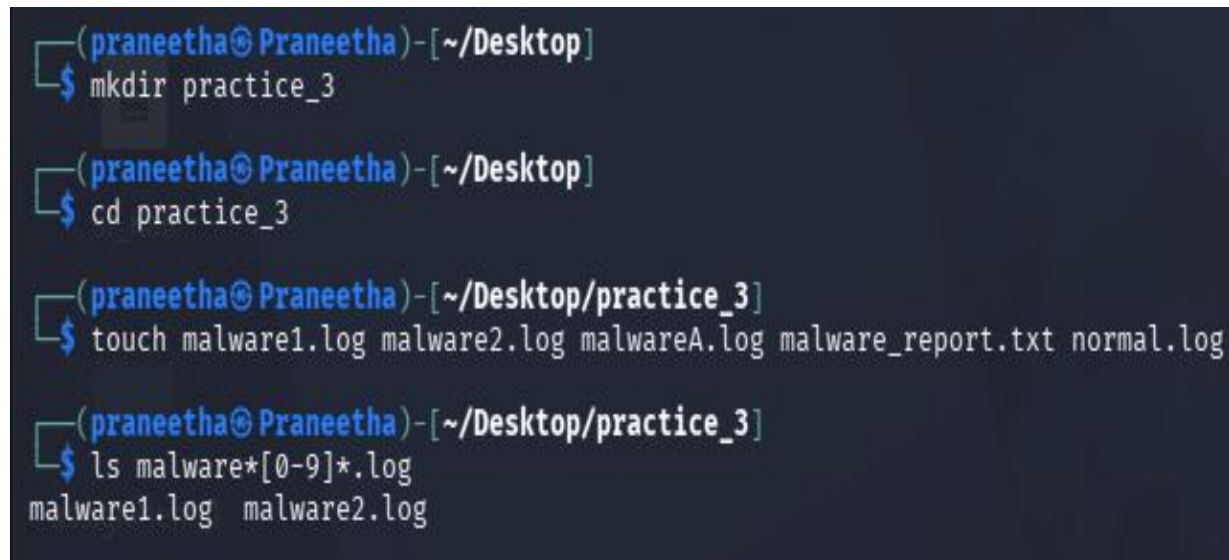
```
(praneetha@Praneetha) - [~/Desktop]
$ mkdir practice_2
(praneetha@Praneetha) - [~/Desktop]
$ cd practice_2
(praneetha@Praneetha) - [~/Desktop/practice_2]
$ touch access.log error.log auth.log auth.log.1 auth.log.2 system.log notes.txt
(praneetha@Praneetha) - [~/Desktop/practice_2]
$ ls @(access|error|auth).log
access.log auth.log error.log
(praneetha@Praneetha) - [~/Desktop/practice_2]
$ ls !(notes.txt)
access.log auth.log auth.log.1 auth.log.2 error.log system.log
(praneetha@Praneetha) - [~/Desktop/practice_2]
$ ls auth.log.@(1|2)
auth.log.1 auth.log.2
```


b. Given the files:

malware1.log, malware2.log, malwareA.log, malware_report.txt, normal.log

construct a glob pattern that identifies .log files whose names start with malware and contain a numeric character.

```
ls malware*[0-9]*.log
```



```
(praneetha@Praneetha) - [~/Desktop]
$ mkdir practice_3

(praneetha@Praneetha) - [~/Desktop]
$ cd practice_3

(praneetha@Praneetha) - [~/Desktop/practice_3]
$ touch malware1.log malware2.log malwareA.log malware_report.txt normal.log

(praneetha@Praneetha) - [~/Desktop/practice_3]
$ ls malware*[0-9]*.log
malware1.log  malware2.log
```

c. Given:

failed_login.log, successful_login.log, login_backup.log, login.txt, firewall.log

- use extended globbing to select .log files containing either failed_login or successful_login.

```
ls @(failed_login|successful_login).log
```

- Construct a pattern that lists all .log files except firewall.log.

```
ls !(*firewall*).log
```

```
(praneetha@Praneetha)-[~/Desktop]
$ mkdir practice_4

(praneetha@Praneetha)-[~/Desktop]
$ cd practice_4

(praneetha@Praneetha)-[~/Desktop/practice_4]
$ touch failed_login.log successful_login.log login_backup.log login.txt firewall.log

(praneetha@Praneetha)-[~/Desktop/practice_4]
$ ls @(failed_login|successful_login).log
failed_login.log  successful_login.log

(praneetha@Praneetha)-[~/Desktop/practice_4]
$ ls !(*firewall*).log
failed_login.log  login_backup.log  successful_login.log
```

7. Filters

- a. Display the first 10 entries of login.log.

```
(praneetha@Praneetha)~[/Desktop/Logs]
$ head -n 10 login.log
Aug 15 09:00:00 server sshd[2101]: Accepted password for student1 from 192.168.10.21
Aug 15 09:01:07 server sshd[2102]: Failed password for student2 from 198.51.100.44
Aug 15 09:02:14 server sshd[2103]: Failed password for analyst from 203.0.113.77
Aug 15 09:03:21 server sshd[2104]: Accepted password for admin from 192.168.10.45
Aug 15 09:04:28 server sshd[2105]: Accepted password for root from 192.168.10.21
Aug 15 09:05:35 server sshd[2106]: Failed password for guest from 198.51.100.44
Aug 15 09:06:42 server sshd[2107]: Failed password for student1 from 203.0.113.77
Aug 15 09:07:49 server sshd[2108]: Accepted password for student2 from 192.168.10.45
Aug 15 09:08:56 server sshd[2109]: Accepted password for analyst from 192.168.10.21
Aug 15 09:09:03 server sshd[2110]: Failed password for admin from 198.51.100.44
```

- b. Display the first 5 entries of access.log.

```
(praneetha@Praneetha)~[/Desktop/Logs]
$ head -n 5 access.log
192.168.10.21 - - [15/Aug/2026:09:00:00 +0530] "GET / HTTP/1.1" 200 512
192.168.10.22 - - [15/Aug/2026:09:00:37 +0530] "POST /login HTTP/1.1" 200 649
192.168.10.30 - - [15/Aug/2026:09:01:14 +0530] "GET /dashboard HTTP/1.1" 302 786
192.168.10.45 - - [15/Aug/2026:09:01:51 +0530] "GET /admin HTTP/1.1" 403 923
203.0.113.25 - - [15/Aug/2026:09:02:28 +0530] "GET /robots.txt HTTP/1.1" 404 1060
```

- c. Display the first 15 security events from security.log.

```
(praneetha@Praneetha)~[/Desktop/Logs]
$ head -n 15 security.log
2026-08-15 09:00:00 INFO user login successful; source=192.168.10.21; event_id=SEC0001
2026-08-15 09:00:29 WARN failed login attempt; source=192.168.10.22; event_id=SEC0002
2026-08-15 09:00:58 ALERT firewall blocked connection; source=192.168.10.30; event_id=SEC0003
2026-08-15 09:01:27 INFO antivirus scan completed; source=192.168.10.45; event_id=SEC0004
2026-08-15 09:01:56 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0005
2026-08-15 09:02:25 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0006
2026-08-15 09:02:54 INFO user logout; source=203.0.113.77; event_id=SEC0007
2026-08-15 09:03:23 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0008
2026-08-15 09:03:52 INFO user login successful; source=192.168.10.21; event_id=SEC0009
2026-08-15 09:04:21 WARN failed login attempt; source=192.168.10.22; event_id=SEC0010
2026-08-15 09:04:50 ALERT firewall blocked connection; source=192.168.10.30; event_id=SEC0011
2026-08-15 09:05:19 INFO antivirus scan completed; source=192.168.10.45; event_id=SEC0012
2026-08-15 09:05:48 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0013
2026-08-15 09:06:17 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0014
2026-08-15 09:06:46 INFO user logout; source=203.0.113.77; event_id=SEC0015
```

- d. Display the first 20 events recorded in incident.log.

```
(praneetha@Praneetha)~[/Desktop/Logs]
$ head -n 20 incident.log
2026-08-15 09:00:00 INFO login success; source=192.168.10.21; incident_event=001
2026-08-15 09:00:13 INFO web request; source=192.168.10.22; incident_event=002
2026-08-15 09:00:26 INFO logout; source=192.168.10.30; incident_event=003
2026-08-15 09:00:39 INFO scheduled scan; source=192.168.10.45; incident_event=004
2026-08-15 09:00:52 INFO access check; source=203.0.113.25; incident_event=005
2026-08-15 09:01:05 INFO login success; source=198.51.100.44; incident_event=006
2026-08-15 09:01:18 INFO web request; source=203.0.113.77; incident_event=007
2026-08-15 09:01:31 INFO logout; source=198.51.100.91; incident_event=008
2026-08-15 09:01:44 INFO scheduled scan; source=192.168.10.21; incident_event=009
2026-08-15 09:01:57 INFO access check; source=192.168.10.22; incident_event=010
2026-08-15 09:02:10 INFO login success; source=192.168.10.30; incident_event=011
2026-08-15 09:02:23 INFO web request; source=192.168.10.45; incident_event=012
2026-08-15 09:02:36 INFO logout; source=203.0.113.25; incident_event=013
2026-08-15 09:02:49 INFO scheduled scan; source=198.51.100.44; incident_event=014
2026-08-15 09:03:02 INFO access check; source=203.0.113.77; incident_event=015
2026-08-15 09:03:15 INFO login success; source=198.51.100.91; incident_event=016
2026-08-15 09:03:28 INFO web request; source=192.168.10.21; incident_event=017
2026-08-15 09:03:41 INFO logout; source=192.168.10.22; incident_event=018
2026-08-15 09:03:54 INFO scheduled scan; source=192.168.10.30; incident_event=019
2026-08-15 09:04:07 INFO access check; source=192.168.10.45; incident_event=020
```

- e. Display only the first entry in login.log.

```
(praneetha@Praneetha)~[/Desktop/Logs]
$ head -n 1 login.log
Aug 15 09:00:00 server sshd[2101]: Accepted password for student1 from 192.168.10.21
```

- f. Display the first 25 web requests recorded in access.log.

```
(praneetha@Praneetha)~[/Desktop/Logs]
$ head -n 25 access.log
192.168.10.21 - - [15/Aug/2026:09:00:00 +0530] "GET / HTTP/1.1" 200 512
192.168.10.22 - - [15/Aug/2026:09:00:37 +0530] "POST /login HTTP/1.1" 200 649
192.168.10.30 - - [15/Aug/2026:09:01:14 +0530] "GET /dashboard HTTP/1.1" 302 786
192.168.10.45 - - [15/Aug/2026:09:01:51 +0530] "GET /admin HTTP/1.1" 403 923
203.0.113.25 - - [15/Aug/2026:09:02:28 +0530] "GET /robots.txt HTTP/1.1" 404 1060
198.51.100.44 - - [15/Aug/2026:09:03:05 +0530] "POST /api/status HTTP/1.1" 401 1197
203.0.113.77 - - [15/Aug/2026:09:03:42 +0530] "GET /assets/app.js HTTP/1.1" 200 1334
198.51.100.91 - - [15/Aug/2026:09:04:19 +0530] "GET / HTTP/1.1" 200 1471
192.168.10.21 - - [15/Aug/2026:09:04:56 +0530] "GET /login HTTP/1.1" 302 1608
192.168.10.22 - - [15/Aug/2026:09:05:33 +0530] "POST /dashboard HTTP/1.1" 403 1745
192.168.10.30 - - [15/Aug/2026:09:06:10 +0530] "GET /admin HTTP/1.1" 404 1882
192.168.10.45 - - [15/Aug/2026:09:06:47 +0530] "GET /robots.txt HTTP/1.1" 401 2019
203.0.113.25 - - [15/Aug/2026:09:07:24 +0530] "GET /api/status HTTP/1.1" 200 2156
198.51.100.44 - - [15/Aug/2026:09:08:01 +0530] "POST /assets/app.js HTTP/1.1" 200 2293
203.0.113.77 - - [15/Aug/2026:09:08:38 +0530] "GET / HTTP/1.1" 302 2430
198.51.100.91 - - [15/Aug/2026:09:09:15 +0530] "GET /login HTTP/1.1" 403 2567
192.168.10.21 - - [15/Aug/2026:09:09:52 +0530] "GET /dashboard HTTP/1.1" 404 2704
192.168.10.22 - - [15/Aug/2026:09:10:29 +0530] "POST /admin HTTP/1.1" 401 2841
192.168.10.30 - - [15/Aug/2026:09:11:06 +0530] "GET /robots.txt HTTP/1.1" 200 2978
192.168.10.45 - - [15/Aug/2026:09:11:43 +0530] "GET /api/status HTTP/1.1" 200 3115
203.0.113.25 - - [15/Aug/2026:09:12:20 +0530] "GET /assets/app.js HTTP/1.1" 302 3252
198.51.100.44 - - [15/Aug/2026:09:12:57 +0530] "POST / HTTP/1.1" 403 3389
203.0.113.77 - - [15/Aug/2026:09:13:34 +0530] "GET /login HTTP/1.1" 404 3526
198.51.100.91 - - [15/Aug/2026:09:14:11 +0530] "GET /dashboard HTTP/1.1" 401 3663
192.168.10.21 - - [15/Aug/2026:09:14:48 +0530] "GET /admin HTTP/1.1" 200 3800
```

- g. Display the last 10 entries of login.log.

```
(praneetha@Praneetha)-[~/Desktop/Logs]
$ tail -n 10 login.log
Aug 15 09:40:40 server sshd[2141]: Accepted password for root from 192.168.10.21
Aug 15 09:41:47 server sshd[2142]: Failed password for guest from 198.51.100.44
Aug 15 09:42:54 server sshd[2143]: Failed password for student1 from 203.0.113.77
Aug 15 09:43:01 server sshd[2144]: Accepted password for student2 from 192.168.10.45
Aug 15 09:44:08 server sshd[2145]: Accepted password for analyst from 192.168.10.21
Aug 15 09:45:15 server sshd[2146]: Failed password for admin from 198.51.100.44
Aug 15 09:46:22 server sshd[2147]: Failed password for root from 203.0.113.77
Aug 15 09:47:29 server sshd[2148]: Accepted password for guest from 192.168.10.45
Aug 15 09:48:36 server sshd[2149]: Accepted password for student1 from 192.168.10.21
Aug 15 09:49:43 server sshd[2150]: Failed password for student2 from 198.51.100.44
```

- h. Display the last 5 entries of access.log.

```
(praneetha@Praneetha)-[~/Desktop/Logs]
$ tail -n 5 access.log
192.168.10.45 - - [15/Aug/2026:09:46:15 +0530] "GET /api/status HTTP/1.1" 403 2595
203.0.113.25 - - [15/Aug/2026:09:46:52 +0530] "GET /assets/app.js HTTP/1.1" 404 2732
198.51.100.44 - - [15/Aug/2026:09:47:29 +0530] "POST / HTTP/1.1" 401 2869
203.0.113.77 - - [15/Aug/2026:09:48:06 +0530] "GET /login HTTP/1.1" 200 3006
198.51.100.91 - - [15/Aug/2026:09:48:43 +0530] "GET /dashboard HTTP/1.1" 200 3143
```

- i. Display the 20 most recent security events from security.log.

```
(praneetha@Praneetha)-[~/Desktop/Logs]
$ tail -n 20 security.log
2026-08-15 09:48:20 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0101
2026-08-15 09:48:49 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0102
2026-08-15 09:49:18 INFO user logout; source=203.0.113.77; event_id=SEC0103
2026-08-15 09:49:47 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0104
2026-08-15 09:50:16 INFO user login successful; source=192.168.10.21; event_id=SEC0105
2026-08-15 09:50:45 WARN failed login attempt; source=192.168.10.22; event_id=SEC0106
2026-08-15 09:51:14 ALERT firewall blocked connection; source=192.168.10.30; event_id=SEC0107
2026-08-15 09:51:43 INFO antivirus scan completed; source=192.168.10.45; event_id=SEC0108
2026-08-15 09:52:12 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0109
2026-08-15 09:52:41 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0110
2026-08-15 09:53:10 INFO user logout; source=203.0.113.77; event_id=SEC0111
2026-08-15 09:53:39 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0112
2026-08-15 09:54:08 INFO user login successful; source=192.168.10.21; event_id=SEC0113
2026-08-15 09:54:37 WARN failed login attempt; source=192.168.10.22; event_id=SEC0114
2026-08-15 09:55:06 ALERT firewall blocked connection; source=192.168.10.30; event_id=SEC0115
2026-08-15 09:55:35 INFO antivirus scan completed; source=192.168.10.45; event_id=SEC0116
2026-08-15 09:56:04 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0117
2026-08-15 09:56:33 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0118
2026-08-15 09:57:02 INFO user logout; source=203.0.113.77; event_id=SEC0119
2026-08-15 09:57:31 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0120
```


- j. Display the 25 most recent events from incident.log.

```
(praneetha@Praneetha) - [~/Desktop/Logs]
$ tail -n 25 incident.log
2026-08-15 10:42:55 INFO login success; source=192.168.10.45; incident_event=476
2026-08-15 10:43:08 INFO web request; source=203.0.113.25; incident_event=477
2026-08-15 10:43:21 INFO logout; source=198.51.100.44; incident_event=478
2026-08-15 10:43:34 INFO scheduled scan; source=203.0.113.77; incident_event=479
2026-08-15 10:43:47 INFO access check; source=198.51.100.91; incident_event=480
2026-08-15 10:44:00 INFO login success; source=192.168.10.21; incident_event=481
2026-08-15 10:44:13 INFO web request; source=192.168.10.22; incident_event=482
2026-08-15 10:44:26 INFO logout; source=192.168.10.30; incident_event=483
2026-08-15 10:44:39 INFO scheduled scan; source=192.168.10.45; incident_event=484
2026-08-15 10:44:52 INFO access check; source=203.0.113.25; incident_event=485
2026-08-15 10:45:05 INFO login success; source=198.51.100.44; incident_event=486
2026-08-15 10:45:18 INFO web request; source=203.0.113.77; incident_event=487
2026-08-15 10:45:31 INFO logout; source=198.51.100.91; incident_event=488
2026-08-15 10:45:44 INFO scheduled scan; source=192.168.10.21; incident_event=489
2026-08-15 10:45:57 INFO access check; source=192.168.10.22; incident_event=490
2026-08-15 10:46:10 INFO login success; source=192.168.10.30; incident_event=491
2026-08-15 10:46:23 INFO web request; source=192.168.10.45; incident_event=492
2026-08-15 10:46:36 INFO logout; source=203.0.113.25; incident_event=493
2026-08-15 10:46:49 INFO scheduled scan; source=198.51.100.44; incident_event=494
2026-08-15 10:47:02 INFO access check; source=203.0.113.77; incident_event=495
2026-08-15 10:47:15 INFO login success; source=198.51.100.91; incident_event=496
2026-08-15 10:47:28 INFO web request; source=192.168.10.21; incident_event=497
2026-08-15 10:47:41 INFO logout; source=192.168.10.22; incident_event=498
2026-08-15 10:47:54 INFO scheduled scan; source=192.168.10.30; incident_event=499
2026-08-15 10:48:07 INFO access check; source=192.168.10.45; incident_event=500
```

- k. Display only the last entry in security.log.

```
(praneetha@Praneetha) - [~/Desktop/Logs]
$ tail -n 1 security.log
2026-08-15 09:57:31 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0120
```

- l. Display the last 50 events recorded in incident.log.

```
(praneetha@Praneetha) - [~/Desktop/Logs]
$ tail -n 50 incident.log
2026-08-15 10:37:30 INFO login success; source=192.168.10.30; incident_event=451
2026-08-15 10:37:43 INFO web request; source=192.168.10.45; incident_event=452
2026-08-15 10:37:56 INFO logout; source=203.0.113.25; incident_event=453
2026-08-15 10:38:09 INFO scheduled scan; source=198.51.100.44; incident_event=454
2026-08-15 10:38:22 INFO access check; source=203.0.113.77; incident_event=455
2026-08-15 10:38:35 INFO login success; source=198.51.100.91; incident_event=456
2026-08-15 10:38:48 INFO web request; source=192.168.10.21; incident_event=457
2026-08-15 10:39:01 INFO logout; source=192.168.10.22; incident_event=458
2026-08-15 10:39:14 INFO scheduled scan; source=192.168.10.30; incident_event=459
2026-08-15 10:39:27 INFO access check; source=192.168.10.45; incident_event=460
2026-08-15 10:39:40 INFO login success; source=203.0.113.25; incident_event=461
2026-08-15 10:39:53 INFO web request; source=198.51.100.44; incident_event=462
2026-08-15 10:40:06 INFO logout; source=203.0.113.77; incident_event=463
```

- m. Display the contents of login.log starting from line 41 until the end of the file.

```
(praneetha@Praneetha)~[~/Desktop/Logs]
$ tail -n +41 login.log
Aug 15 09:40:40 server sshd[2141]: Accepted password for root from 192.168.10.21 login.log
Aug 15 09:41:47 server sshd[2142]: Failed password for guest from 198.51.100.44 access.log
Aug 15 09:42:54 server sshd[2143]: Failed password for student1 from 203.0.113.77
Aug 15 09:43:01 server sshd[2144]: Accepted password for student2 from 192.168.10.45
Aug 15 09:44:08 server sshd[2145]: Accepted password for analyst from 192.168.10.21
Aug 15 09:45:15 server sshd[2146]: Failed password for admin from 198.51.100.44
Aug 15 09:46:22 server sshd[2147]: Failed password for root from 203.0.113.77
Aug 15 09:47:29 server sshd[2148]: Accepted password for guest from 192.168.10.45 wc access.log
Aug 15 09:48:36 server sshd[2149]: Accepted password for student1 from 192.168.10.21 n.log access.log
Aug 15 09:49:43 server sshd[2150]: Failed password for student2 from 198.51.100.44
```

- n. Display the contents of security.log starting from line 101 until the end of the file.

```
(praneetha@Praneetha)~[~/Desktop/Logs]
$ tail -n +101 security.log
2026-08-15 09:48:20 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0101
2026-08-15 09:48:49 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0102
2026-08-15 09:49:18 INFO user logout; source=203.0.113.77; event_id=SEC0103
2026-08-15 09:49:47 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0104
2026-08-15 09:50:16 INFO user login successful; source=192.168.10.21; event_id=SEC0105
2026-08-15 09:50:45 WARN failed login attempt; source=192.168.10.22; event_id=SEC0106
2026-08-15 09:51:14 ALERT firewall blocked connection; source=192.168.10.30; event_id=SEC0107
2026-08-15 09:51:43 INFO antivirus scan completed; source=192.168.10.45; event_id=SEC0108
2026-08-15 09:52:12 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0109
2026-08-15 09:52:41 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0110
2026-08-15 09:53:10 INFO user logout; source=203.0.113.77; event_id=SEC0111
2026-08-15 09:53:39 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0112
```

- o. Find the total number of lines in login.log.

```
(praneetha@Praneetha)~[~/Desktop/Logs]
$ wc -l login.log
50 login.log
```

- p. Find the total number of lines in access.log.

```
(praneetha@Praneetha)~[~/Desktop/Logs]
$ wc -l access.log
80 access.log
```

- q. Find the total number of security events in security.log.

```
(praneetha@Praneetha)-[~/Desktop/Logs]  
$ wc -l security.log  
120 security.log
```

- r. Determine the total number of events recorded in incident.log.

```
(praneetha@Praneetha)-[~/Desktop/Logs]  
$ wc -l incident.log  
500 incident.log
```

- s. Count the number of words in login.log.

```
(praneetha@Praneetha)-[~/Desktop/Logs]  
$ wc -w login.log  
550 login.log
```

- t. Count the number of bytes in security.log.

```
(praneetha@Praneetha)-[~/Desktop/Logs]  
$ wc -c security.log  
10875 security.log
```

- u. Display the line, word, and byte counts of access.log.

```
(praneetha@Praneetha)-[~/Desktop/Logs]  
$ wc access.log  
80 800 6414 access.log
```


- v. Find the line counts of all four .log files using a single wc command.

```
(praneetha@Praneetha)-[~/Desktop/Logs]
$ wc -l login.log access.log security.log incident.log
 50 login.log
 80 access.log
120 security.log
500 incident.log
750 total
```

- w. Display the first 20 entries of login.log and, from those entries, display only the last 5.

```
(praneetha@Praneetha)-[~/Desktop/Logs]
$ head -n 20 login.log | tail -n 5 login.log
Aug 15 09:15:45 server sshd[2116]: Accepted password for admin from 192.168.10.45
Aug 15 09:16:52 server sshd[2117]: Accepted password for root from 192.168.10.21
Aug 15 09:17:59 server sshd[2118]: Failed password for guest from 198.51.100.44
Aug 15 09:18:06 server sshd[2119]: Failed password for student1 from 203.0.113.77
Aug 15 09:19:13 server sshd[2120]: Accepted password for student2 from 192.168.10.45
```

- x. Display the last 20 entries of login.log and, from those entries, display only the first 5.

```
(praneetha@Praneetha)-[~/Desktop/Logs]
$ tail -n 20 login.log | head -n 5
Aug 15 09:30:30 server sshd[2131]: Failed password for student1 from 203.0.113.77
Aug 15 09:31:37 server sshd[2132]: Accepted password for student2 from 192.168.10.45
Aug 15 09:32:44 server sshd[2133]: Accepted password for analyst from 192.168.10.21
Aug 15 09:33:51 server sshd[2134]: Failed password for admin from 198.51.100.44
Aug 15 09:34:58 server sshd[2135]: Failed password for root from 203.0.113.77
```

- y. Display the first 30 entries of access.log and then display only the last 10 entries from that output.

```
(praneetha@Praneetha)-[~/Desktop/Logs]
$ head -n 30 access.log | tail -n 10
203.0.113.25 - - [15/Aug/2026:09:12:20 +0530] "GET /assets/app.js HTTP/1.1" 302 3252
198.51.100.44 - - [15/Aug/2026:09:12:57 +0530] "POST / HTTP/1.1" 403 3389
203.0.113.77 - - [15/Aug/2026:09:13:34 +0530] "GET /login HTTP/1.1" 404 3526
198.51.100.91 - - [15/Aug/2026:09:14:11 +0530] "GET /dashboard HTTP/1.1" 401 3663
192.168.10.21 - - [15/Aug/2026:09:14:48 +0530] "GET /admin HTTP/1.1" 200 3800
192.168.10.22 - - [15/Aug/2026:09:15:25 +0530] "POST /robots.txt HTTP/1.1" 200 3937
192.168.10.30 - - [15/Aug/2026:09:16:02 +0530] "GET /api/status HTTP/1.1" 302 4074
192.168.10.45 - - [15/Aug/2026:09:16:39 +0530] "GET /assets/app.js HTTP/1.1" 403 4211
203.0.113.25 - - [15/Aug/2026:09:17:16 +0530] "GET / HTTP/1.1" 404 4348
198.51.100.44 - - [15/Aug/2026:09:17:53 +0530] "POST /login HTTP/1.1" 401 4485
```

- z. Display the last 30 entries of access.log and then display only the first 10 entries.

```
(praneetha@Praneetha)-[~/Desktop/Logs]
$ tail -n 30 access.log | head -n 10
192.168.10.30 - - [15/Aug/2026:09:30:50 +0530] "GET /login HTTP/1.1" 302 3266
192.168.10.45 - - [15/Aug/2026:09:31:27 +0530] "GET /dashboard HTTP/1.1" 403 3403
203.0.113.25 - - [15/Aug/2026:09:32:04 +0530] "GET /admin HTTP/1.1" 404 3540
198.51.100.44 - - [15/Aug/2026:09:32:41 +0530] "POST /robots.txt HTTP/1.1" 401 3677
203.0.113.77 - - [15/Aug/2026:09:33:18 +0530] "GET /api/status HTTP/1.1" 200 3814
198.51.100.91 - - [15/Aug/2026:09:33:55 +0530] "GET /assets/app.js HTTP/1.1" 200 3951
192.168.10.21 - - [15/Aug/2026:09:34:32 +0530] "GET / HTTP/1.1" 302 4088
192.168.10.22 - - [15/Aug/2026:09:35:09 +0530] "POST /login HTTP/1.1" 403 4225
192.168.10.30 - - [15/Aug/2026:09:35:46 +0530] "GET /dashboard HTTP/1.1" 404 4362
192.168.10.45 - - [15/Aug/2026:09:36:23 +0530] "GET /admin HTTP/1.1" 401 4499
```

- aa. Display the first 50 events from security.log and then display the last 10 events from those 50.

```
(praneetha@Praneetha)-[~/Desktop/Logs]
$ head -n 50 security.log | tail -n 10
2026-08-15 09:19:20 INFO user login successful; source=192.168.10.21; event_id=SEC0041
2026-08-15 09:19:49 WARN failed login attempt; source=192.168.10.22; event_id=SEC0042
2026-08-15 09:20:18 ALERT firewall blocked connection; source=192.168.10.30; event_id=SEC0043
2026-08-15 09:20:47 INFO antivirus scan completed; source=192.168.10.45; event_id=SEC0044
2026-08-15 09:21:16 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0045
2026-08-15 09:21:45 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0046
2026-08-15 09:22:14 INFO user logout; source=203.0.113.77; event_id=SEC0047
2026-08-15 09:22:43 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0048
2026-08-15 09:23:12 INFO user login successful; source=192.168.10.21; event_id=SEC0049
2026-08-15 09:23:41 WARN failed login attempt; source=192.168.10.22; event_id=SEC0050
```

bb. Display the last 50 events from security.log and then display the first 10 events from those 50.

```
(praneetha@Praneetha)~[/Desktop/Logs]
$ tail -n 50 security.log | head -n 10
2026-08-15 09:33:50 INFO user logout; source=203.0.113.77; event_id=SEC0071
2026-08-15 09:34:19 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0072
2026-08-15 09:34:48 INFO user login successful; source=192.168.10.21; event_id=SEC0073
2026-08-15 09:35:17 WARN failed login attempt; source=192.168.10.22; event_id=SEC0074
2026-08-15 09:35:46 ALERT firewall blocked connection; source=192.168.10.30; event_id=SEC0075
2026-08-15 09:36:15 INFO antivirus scan completed; source=192.168.10.45; event_id=SEC0076
2026-08-15 09:36:44 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0077
2026-08-15 09:37:13 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0078
2026-08-15 09:37:42 INFO user logout; source=203.0.113.77; event_id=SEC0079
2026-08-15 09:38:11 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0080
```

cc. Display entries from line 11 to line 20 of login.log.

```
(praneetha@Praneetha)~[/Desktop/Logs]
$ head -n 20 login.log | tail -n 10
Aug 15 09:10:10 server sshd[2111]: Failed password for root from 203.0.113.77
Aug 15 09:11:17 server sshd[2112]: Accepted password for guest from 192.168.10.45
Aug 15 09:12:24 server sshd[2113]: Accepted password for student1 from 192.168.10.21
Aug 15 09:13:31 server sshd[2114]: Failed password for student2 from 198.51.100.44
Aug 15 09:14:38 server sshd[2115]: Failed password for analyst from 203.0.113.77
Aug 15 09:15:45 server sshd[2116]: Accepted password for admin from 192.168.10.45
```

dd. Display entries from line 21 to line 30 of login.log.

```
(praneetha@Praneetha)~[/Desktop/Logs]
$ head -n 30 login.log | tail -n 10
Aug 15 09:20:20 server sshd[2121]: Accepted password for analyst from 192.168.10.21
Aug 15 09:21:27 server sshd[2122]: Failed password for admin from 198.51.100.44
Aug 15 09:22:34 server sshd[2123]: Failed password for root from 203.0.113.77
Aug 15 09:23:41 server sshd[2124]: Accepted password for guest from 192.168.10.45
Aug 15 09:24:48 server sshd[2125]: Accepted password for student1 from 192.168.10.21
Aug 15 09:25:55 server sshd[2126]: Failed password for student2 from 198.51.100.44
Aug 15 09:26:02 server sshd[2127]: Failed password for analyst from 203.0.113.77
```

ee. Display entries from line 11 to line 20 of access.log.

```
(praneetha@Praneetha) - [~/Desktop/Logs]
$ head -n 20 access.log | tail -n 10
192.168.10.30 - - [15/Aug/2026:09:06:10 +0530] "GET /admin HTTP/1.1" 404 1882
192.168.10.45 - - [15/Aug/2026:09:06:47 +0530] "GET /robots.txt HTTP/1.1" 401 2019
203.0.113.25 - - [15/Aug/2026:09:07:24 +0530] "GET /api/status HTTP/1.1" 200 2156
198.51.100.44 - - [15/Aug/2026:09:08:01 +0530] "POST /assets/app.js HTTP/1.1" 200 2293
203.0.113.77 - - [15/Aug/2026:09:08:38 +0530] "GET / HTTP/1.1" 302 2430
198.51.100.91 - - [15/Aug/2026:09:09:15 +0530] "GET /login HTTP/1.1" 403 2567
192.168.10.21 - - [15/Aug/2026:09:09:52 +0530] "GET /dashboard HTTP/1.1" 404 2704
192.168.10.22 - - [15/Aug/2026:09:10:29 +0530] "POST /admin HTTP/1.1" 401 2841
```

ff. Display entries from line 31 to line 40 of access.log.

```
(praneetha@Praneetha) - [~/Desktop/Logs]
$ head -n 40 access.log | tail -n 10
203.0.113.77 - - [15/Aug/2026:09:18:30 +0530] "GET /dashboard HTTP/1.1" 200 526
198.51.100.91 - - [15/Aug/2026:09:19:07 +0530] "GET /admin HTTP/1.1" 200 663
192.168.10.21 - - [15/Aug/2026:09:19:44 +0530] "GET /robots.txt HTTP/1.1" 302 800
192.168.10.22 - - [15/Aug/2026:09:20:21 +0530] "POST /api/status HTTP/1.1" 403 937
192.168.10.30 - - [15/Aug/2026:09:20:58 +0530] "GET /assets/app.js HTTP/1.1" 404 1074
192.168.10.45 - - [15/Aug/2026:09:21:35 +0530] "GET / HTTP/1.1" 401 1211
203.0.113.25 - - [15/Aug/2026:09:22:12 +0530] "GET /login HTTP/1.1" 200 1348
198.51.100.44 - - [15/Aug/2026:09:22:49 +0530] "POST /dashboard HTTP/1.1" 200 1485
203.0.113.77 - - [15/Aug/2026:09:23:26 +0530] "GET /admin HTTP/1.1" 302 1622
198.51.100.91 - - [15/Aug/2026:09:24:03 +0530] "GET /robots.txt HTTP/1.1" 403 1759
```

gg. Display security events from line 21 to line 30 of security.log.

```
(praneetha@Praneetha) - [~/Desktop/Logs]
$ head -n 30 security.log | tail -n 10
2026-08-15 09:09:40 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0021
2026-08-15 09:10:09 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0022
2026-08-15 09:10:38 INFO user logout; source=203.0.113.77; event_id=SEC0023
2026-08-15 09:11:07 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0024
2026-08-15 09:11:36 INFO user login successful; source=192.168.10.21; event_id=SEC0025
2026-08-15 09:12:05 WARN failed login attempt; source=192.168.10.22; event_id=SEC0026
2026-08-15 09:12:34 ALERT firewall blocked connection; source=192.168.10.30; event_id=SEC0027
2026-08-15 09:13:03 INFO antivirus scan completed; source=192.168.10.45; event_id=SEC0028
2026-08-15 09:13:32 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0029
```


hh. Display security events from line 51 to line 60 of security.log.

```
(praneetha@Praneetha) [~/Desktop/Logs]
$ head -n 60 security.log | tail -n 10
2026-08-15 09:24:10 ALERT firewall blocked connection; source=192.168.10.30; event_id=SEC0051
2026-08-15 09:24:39 INFO antivirus scan completed; source=192.168.10.45; event_id=SEC0052
2026-08-15 09:25:08 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0053
2026-08-15 09:25:37 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0054
2026-08-15 09:26:06 INFO user logout; source=203.0.113.77; event_id=SEC0055
2026-08-15 09:26:35 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0056
2026-08-15 09:27:04 INFO user login successful; source=192.168.10.21; event_id=SEC0057
2026-08-15 09:27:33 WARN failed login attempt; source=192.168.10.22; event_id=SEC0058
```

ii. Display incident events from line 101 to line 110 of incident.log.

```
(praneetha@Praneetha) [~/Desktop/Logs]
$ head -n 110 incident.log | tail -n 10
2026-08-15 09:21:40 INFO login success; source=203.0.113.25; incident_event=101
2026-08-15 09:21:53 INFO web request; source=198.51.100.44; incident_event=102
2026-08-15 09:22:06 INFO logout; source=203.0.113.77; incident_event=103
2026-08-15 09:22:19 INFO scheduled scan; source=198.51.100.91; incident_event=104
2026-08-15 09:22:32 INFO access check; source=192.168.10.21; incident_event=105
2026-08-15 09:22:45 INFO login success; source=192.168.10.22; incident_event=106
2026-08-15 09:22:58 INFO web request; source=192.168.10.30; incident_event=107
2026-08-15 09:23:11 INFO logout; source=192.168.10.45; incident_event=108
2026-08-15 09:23:24 INFO scheduled scan; source=203.0.113.25; incident_event=109
```

jj. Display incident events from line 201 to line 220 of incident.log.

```
(praneetha@Praneetha) [~/Desktop/Logs]
$ head -n 220 incident.log | tail -n 20
2026-08-15 09:43:20 WARNING failed SSH login; source=192.168.10.21; incident_event=201
2026-08-15 09:43:33 ALERT firewall block; source=192.168.10.22; incident_event=202
2026-08-15 09:43:46 WARNING possible port scan; source=192.168.10.30; incident_event=203
2026-08-15 09:43:59 ALERT authentication failure; source=192.168.10.45; incident_event=204
2026-08-15 09:44:12 WARNING failed SSH login; source=203.0.113.25; incident_event=205
2026-08-15 09:44:25 ALERT firewall block; source=198.51.100.44; incident_event=206
2026-08-15 09:44:38 WARNING possible port scan; source=203.0.113.77; incident_event=207
2026-08-15 09:44:51 ALERT authentication failure; source=198.51.100.91; incident_event=208
2026-08-15 09:45:04 WARNING failed SSH login; source=192.168.10.21; incident_event=209
2026-08-15 09:45:17 ALERT firewall block; source=192.168.10.22; incident_event=210
2026-08-15 09:45:30 WARNING possible port scan; source=192.168.10.30; incident_event=211
2026-08-15 09:45:43 ALERT authentication failure; source=192.168.10.45; incident_event=212
2026-08-15 09:45:56 WARNING failed SSH login; source=203.0.113.25; incident_event=213
```

kk. Display incident events from line 491 to line 500.

```
(praneetha@Praneetha)-[~/Desktop/Logs]
$ tail -n 10 incident.log
2026-08-15 10:46:10 INFO login success; source=192.168.10.30; incident_event=491
2026-08-15 10:46:23 INFO web request; source=192.168.10.45; incident_event=492
2026-08-15 10:46:36 INFO logout; source=203.0.113.25; incident_event=493
2026-08-15 10:46:49 INFO scheduled scan; source=198.51.100.44; incident_event=494
2026-08-15 10:47:02 INFO access check; source=203.0.113.77; incident_event=495
2026-08-15 10:47:15 INFO login success; source=198.51.100.91; incident_event=496
2026-08-15 10:47:28 INFO web request; source=192.168.10.21; incident_event=497
2026-08-15 10:47:41 INFO logout; source=192.168.10.22; incident_event=498
```

ll. Take the first 20 entries of login.log and count how many lines are displayed.

```
(praneetha@Praneetha)-[~/Desktop/Logs]
$ head -n 20 login.log | wc -l
20
```

mm. Take the last 15 entries of access.log and count how many lines are displayed.

```
(praneetha@Praneetha)-[~/Desktop/Logs]
$ tail -n 15 access.log | wc -l
15
```

nn. Take the first 50 entries of security.log and count the number of words in those entries.

```
(praneetha@Praneetha)-[~/Desktop/Logs]
$ head -n 50 security.log | wc -w
412
```

oo. Take the last 25 entries of security.log and count the number of lines.

```
(praneetha@Praneetha)-[~/Desktop/Logs]
$ tail -n 25 security.log | wc -l
25
```

pp. Extract lines 21–30 from login.log and count the number of extracted lines.

```
(praneetha@Praneetha) - [~/Desktop/Logs]
$ head -n 30 login.log | tail -n 10 | wc -l
10
```

qq. Extract lines 51–60 from access.log and count the number of extracted lines.

```
(praneetha@Praneetha) - [~/Desktop/Logs]
$ head -n 60 access.log | tail -n 10 | wc -l
10
```

rr. Extract events 101–150 from incident.log and count how many events are displayed.

```
(praneetha@Praneetha) - [~/Desktop/Logs]
$ head -n 150 incident.log | tail -n 50 | wc -l
50
```

ss. Display the first 100 events from incident.log, take the last 20 events from that output, and count the number of lines.

```
(praneetha@Praneetha) - [~/Desktop/Logs]
$ head -n 100 incident.log | tail -n 20 | wc -l
20
```